9. Implement a program using Anonymous Functions - Lambda and Filter.

```
# Example 1: Using map() with a function
def square(x):
   return x ** 2
numbers = [1, 2, 3, 4]
squared_numbers = map(square, numbers)
print(list(squared_numbers))

# Example 2: Using map() with lambda
numbers = [1, 2, 3, 4]
squared_numbers = map(lambda x: x ** 2, numbers)
print(list(squared_numbers))

# Example 3: Using filter() with a function
def is_even(x):
   return x % 2 == 0
numbers = [1, 2, 3, 4, 5, 6]
even_numbers = filter(is_even, numbers)
print(list(even_numbers))
```

Output:

```
[1, 4, 9, 16]
[1, 4, 9, 16]
[2, 4, 6]
```

10. Implement a program using Maps, List Comprehension, Dictionaries.

```
# Simple Age Conversion Program

def convert_ages_to_months(names, ages):
    # Using map to convert ages to months
    ages_in_months = list(map(lambda age: age * 12, ages))

    # Creating a dictionary using dictionary comprehension
    name_age_dict = {
        name: age for name, age in zip(names, ages_in_months)
    }

    return name_age_dict


def main():
    # Input: List of names and corresponding ages
    names = ['Alice', 'Bob', 'Charlie', 'Diana']
    ages = [25, 30, 22, 28]

    # Convert ages to months and create the dictionary
    name_age_dict = convert_ages_to_months(names, ages)

    # Output the results
    print("Name to Age in Months Conversion:")

    for name, age_months in name_age_dict.items():
        print(f"{name}: {age_months} months")


# Run the main function
if __name__ == "__main__":
    main()
```

Output:

```
Name to Age in Months Conversion:
Alice: 300 months
Bob: 360 months
Charlie: 264 months
Diana: 336 months
```

11. Implement a program for Class and Object.

```
class Person:
    def __init__(self, name, age):
        self.name = name
        self.age = age

    def display(self):
        print("Name:", self.name)
        print("Age:", self.age)

    def birthday(self):
        self.age = self.age + 1
        print("Happy Birthday,", self.name)
        print("Your age is now:", self.age)

person1 = Person("Sanvi", 18)
person2 = Person("Nilansh", 19)

person1.display()
print()

person2.display()
print()

person1.birthday()
person2.birthday()
```

Output:

```
Name: Sanvi
Age: 18

Name: Nilansh
Age: 19

Happy Birthday, Sanvi
Your age is now: 19
Happy Birthday, Nilansh
Your age is now: 20
```

12. Demonstrate a program using Array in python.

```
def list_example():
    # Create a list (array) of integers
    numbers = [10, 20, 30, 40, 50]

    # Accessing elements
    print("Original list:", numbers)
    print("Element at index 2:", numbers[2])

    numbers[2] = 35
    print("Modified list:", numbers)

    # Appending a new element
    numbers.append(60)
    print("List after appending 60:", numbers)

    # Removing an element
    numbers.remove(20)
    print("List after removing 20:", numbers)

    print("Iterating over the list:")
    for num in numbers:
        print(num)

# Call the function
list_example()
```

Output:

```
Original list: [10, 20, 30, 40, 50]

Element at index 2: 30
Modified list: [10, 20, 35, 40, 50]

List after appending 60: [10, 20, 35, 40, 50, 60]
List after removing 20: [10, 35, 40, 50, 60]

Iterating over the list:
10
35
40
50
60
```

13. Implement a program to Create, Import and use Package and Modules.

1. **Create a directory called** my_package1**.**
   !mkdir -p my_package1

2. **Inside this directory, create a file called __init__.py. This can be empty or contain initialization code. It marks the directory as a package.**
   %%writefile my_package1/__init__.py
   # __init__.py

3. **Move the math_operations.py module into the my_package1 directory.**
   %%writefile my_package1/math_operations.py
   # math_operations.py

```
def add(a, b):
  return a + b

def subtract(a, b):
  return a - b

def multiply(a, b):
  return a * b

def divide(a, b):
  if b == 0:
    raise ValueError("Cannot divide by zero")
  return a / b
```

4. **# main.py**

```
# main.py
from my_package1 import math_operations

# Using the functions from math_operations module
a = 10
b = 5

print(f"{a} + {b} = {math_operations.add(a, b)}")
print(f"{a} - {b} = {math_operations.subtract(a, b)}")
print(f"{a} * {b} = {math_operations.multiply(a, b)}")
print(f"{a} / {b} = {math_operations.divide(a, b)}")
```

Output:

```
10 + 5 = 15
10 - 5 = 5
10 * 5 = 50
10 / 5 = 2.0
```

In [1]:
```
!mkdir my_package1
```

In [13]:
```
%%writefile my_package1/__init__.py
# __init__.py can be empty or contain initialization code
```

```
Overwriting my_package1/__init__.py
```

In [14]:
```
%%writefile my_package1/math_operations.py
# math_operations.py
def add(a, b):
    return a + b

def subtract(a, b):
    return a - b

def multiply(a, b):
    return a * b

def divide(a, b):
    if b == 0:
        raise ValueError("Cannot divide by zero")
    return a / b
```

```
Writing my_package1/math_operations.py
```

In [15]:
```
# main.py
from my_package1 import math_operations

# Using the functions from math_operations module
a = 10
b = 5

print(f"{a} + {b} = {math_operations.add(a, b)}")
print(f"{a} - {b} = {math_operations.subtract(a, b)}")
print(f"{a} * {b} = {math_operations.multiply(a, b)}")
print(f"{a} / {b} = {math_operations.divide(a, b)}")

```

```
10 + 5 = 15
10 - 5 = 5
10 * 5 = 50
10 / 5 = 2.0
```

This example demonstrates how to create, import, and use a package and modules in Python. The package my_package contains the module math_operations, which is imported and used in the main.py script.

Another Program:

File 1) mymodule.py person1
= {
"name": "John",
"age": 36,
"country": "Norway"
}

File 2) File2.py

```
import mymodule
#mymodule is the first file name
a = mymodule.person1["age"]
print(a)
```

Note: Run File2.py (save File 1 as a name mymodule.py and this name use as a package in File 2)